Supplementary Material for "ExactFuse: Closed-Form Unfolding With Exact Operators for Sensor-Independent Hyperspectral and Multispectral Image Fusion"

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This document accompanies the main paper. Numbers with the prefix S refer to this document; all others refer to the main paper. The result files behind every table and figure are in the public repository (https://github.com/Amarnath10i/Research-On-Hyperspectral-ImageFussion).

S-I. Training Stages and the Exact Border

Chikusei. The final Chikusei model was obtained in two runs: a first run from scratch with zero-padded operators and the L1 losses only, and a fine-tuning run with the exact reflect-border operators and the full loss. Table S-I traces the path from the first run to the final model. Evaluating the first-run weights with the exact reflect-border operator, without retraining, drops the validation PSNR from 58.60 to 49.52 dB. The prior had learned to undo the error that the zero-padded operator introduces at the border, so with the exact operator it over-corrects there; the effect is strongest on the small 64×64 validation patches, where border pixels are a large fraction of the image. Fine-tuning removes this within 2000 iterations (58.60 dB validation) and ends at 58.72 dB. The consistency of the reconstruction with its inputs rises from 76.7 to 85.5 dB for the LR-HSI and from 75.0 to 89.0 dB for the HR-MSI. Fig. S1 shows the validation curves; validation PSNR stops improving after about 44,000 fine-tuning iterations, and the test PSNR stays between 58.13 and 58.15 dB from iteration 26,000 onward.

Pavia Centre. The Pavia Centre model was trained from scratch with the exact operators and the full loss. Validation PSNR stops improving after 43,000 iterations (Fig. S2), and those weights are tested; the test PSNR, logged but never used for selection, peaks at 50.66 dB at 25,000 iterations, against 50.63 dB for the selected weights. The reconstruction reproduces its inputs to within 79.9 dB (LR-HSI) and 77.6 dB (HR-MSI).

TABLE S-I
From the first run to the final Chikusei model (test set unless marked). Cons.: PSNR($\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_{\mathrm{H}}$) / PSNR($\mathcal{R}\hat{\mathbf{X}}, \mathbf{Y}_{\mathrm{M}}$) in dB; –: not evaluated

Model	Val.	PSNR	SAM	ERGAS	Cons.
GSA	–	43.20	2.730	3.279	45.4 / 45.0
First run (zero border)	58.60	57.99	0.728	1.356	76.7 / 75.0
+ self-ensemble	–	58.02	0.726	1.353	76.8 / 75.0
First-run weights, exact border	49.52	54.24	0.773	1.474	–
Fine-tuned (exact border, full loss)	58.72	58.15	0.717	1.337	85.5 / 89.0
+ self-ensemble	–	58.20	0.713	1.330	86.0 / 89.5

S-II. Robustness to Degradation Mismatch and Noise

Published methods are usually trained and tested with the same degradation. We test the final models on inputs made with a different blur or a shifted spectral response, and with Gaussian noise added to both observations (Table S-II, Fig. S3). Because the operators are explicit, ExactFuse can also be given the true test-time operator without retraining ("swap"). On Chikusei, a blur mismatch costs 3.9–7.5 dB with the training operator. With the true blur swapped in, the loss is at most 0.26 dB for σ between 1.5 and 3.0; with the zero-border operators of the first run, the same swap recovered only to 55.2–57.1 dB. A shifted spectral response is not compensated by the swap, which lowers the PSNR by a further 0.5 dB: the priors, trained with one response, rely on it beyond the data step. Noise is the main weakness of a model trained on noise-free data; ExactFuse still exceeds GSA by 10.0 dB at 40 dB SNR and by 2.8 dB at 30 dB SNR. On Pavia Centre the blur behaves the same way: the training operator loses 3.3–6.8 dB and the swapped-in true operator at most 0.11 dB. Unlike on Chikusei, swapping in a shifted IKONOS response raises the PSNR slightly (by 0.2–0.3 dB), perhaps because an 8 nm shift is small relative to the 66–96 nm IKONOS bands, and the margins over GSA under noise are larger (12.8, 9.8 and 6.0 dB at 40, 35 and 30 dB SNR).

S-III. Q2n on Chikusei

Q2n normalizes every band of every 32×32 block by the ground truth's own mean and standard deviation in that block, so an error that is small in absolute terms counts heavily in a band that is nearly flat within a block. Table S-III and Fig. S4 decompose the Q2n of the final model (0.9931). If the eight ultraviolet bands (363–404 nm) were reconstructed exactly, Q2n would rise to 0.9988; if the last 24 bands were exact as well, to 0.9998. In the worst 10% of blocks, the error energy of these eight bands equals 0.41 of the ground truth's local variance, against at most 0.040 for any other group of eight bands. The fine detail of these bands cannot be predicted from the inputs: a least-squares injection of the multispectral detail, fitted on the test ground truth itself, reduces the band-1 error only from 61.6 to 57.5 DN, while ExactFuse reaches 51.8 DN,

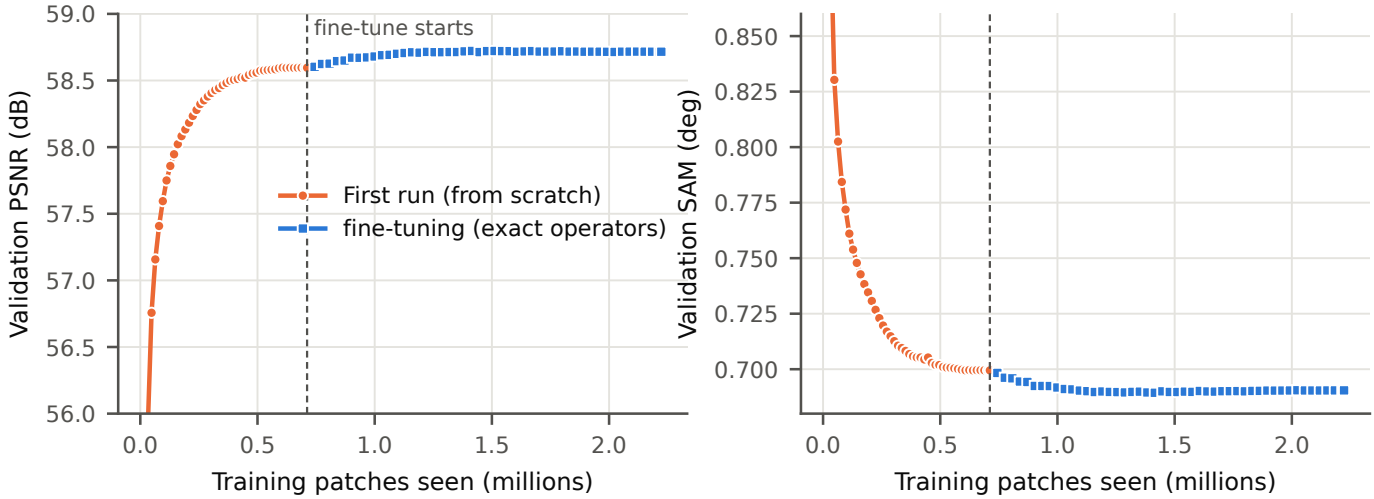


Fig. S1. Chikusei: validation PSNR and SAM during training, plotted against the number of 64×64 training patches seen. The first run starts from random weights (its first two evaluations lie below the plotted range); the fine-tuning run continues from its final weights with the exact border and the full loss.

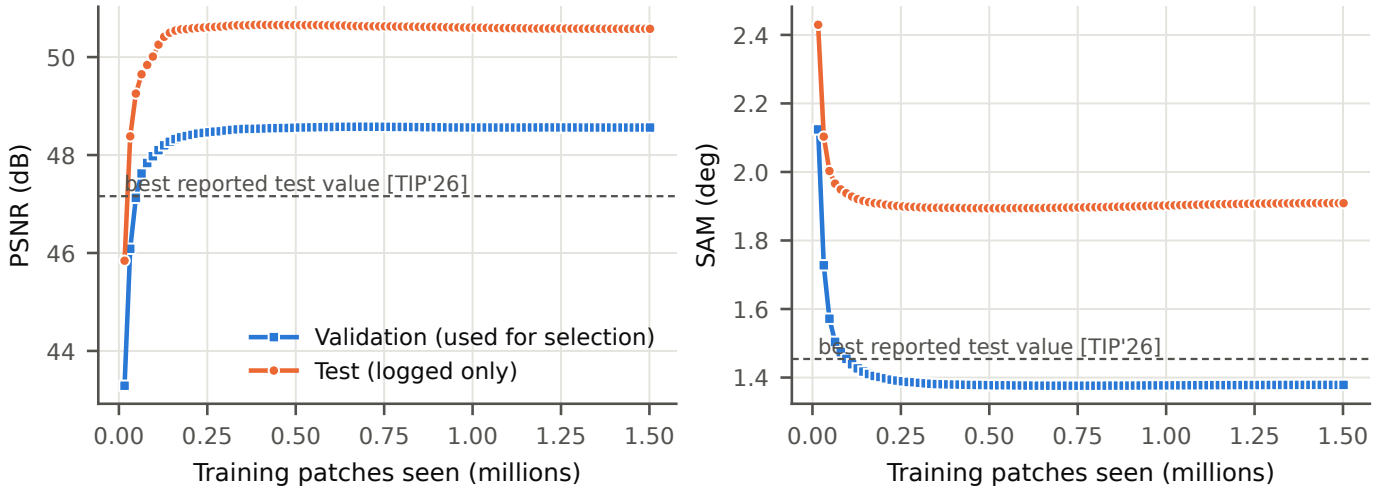


Fig. S2. Pavia Centre: PSNR and SAM during training from scratch, on the validation patches (used to select the weights) and on the test images (logged only). The dashed line is the best value reported in [1] on the PSRT benchmark data.

and ExactFuse is more accurate than this oracle in every band we checked.

Q2n is therefore nearly insensitive to overall accuracy on these test images: blending our reconstruction toward bicubic interpolation lowers the PSNR from 58.20 to 52.15 dB but Q2n only from 0.9931 to 0.9904 (Fig. S4, right). The values reported in [1] do not follow PSNR in this way. Between 52.2 and 53.6 dB they range from 0.9832 (DCT) to 0.9963 (DDIF); of the 16 methods within the PSNR range of our curve, 11 lie above it and 5 below. U2Net, for example, is listed at 0.9928 at 50.55 dB, a value our implementation reaches only near 57.6 dB. We conclude that the Q2n values of [1] were most likely computed with a different implementation, normalization or test region, and that they are not directly comparable with ours.

S-IV. SAM and Q2n on Pavia Centre

The Q2n shortfall on Pavia Centre comes from one test image, which also has the largest SAM (Table S-V, Fig. S5). On images 1, 3 and 4, Q2n is 0.9987–0.9991, at or above the best reported value, but SAM is still 1.55–1.68° (mean 1.60°), above the best reported 1.454°: the water image explains the Q2n gap, but only part of the SAM gap. Image 2 contains a stretch of the river Ticino: 63% of its pixels have a positive normalized difference water index (green against near infrared), against 11, 7 and 13% in the other images, and its SAM is 2.72° and its Q2n 0.904. Water reflects little light, so its spectra are dominated by sensor noise that neither observation carries. The per-band noise that we estimate from the ground truth alone makes an angle of 3.8° with the noise-free spectrum in the darkest tenth of the pixels, against 0.23° in the brightest tenth. Over water ExactFuse's SAM is 2.95°, over all other pixels 1.56°, and the darkest 10% of the pixels account for 34% of its mean SAM. The same pixels make image 2 nearly flat within many 32×32 blocks, where Q2n divides the error by a very small local variance; image 2 alone carries 97% of the Q2n deficit $\sum_i (1 - \mathrm{Q2n}_i)$ of the four images.

TABLE S-II
ROBUSTNESS OF THE FINAL MODELS (TEST PSNR IN DB / SAM IN DEGREES). FIXED: TRAINING OPERATOR; SWAP: TRUE OPERATOR SUPPLIED AT TEST TIME; –: NOT APPLICABLE. THE BETTER BASE-MODEL ENTRY OF EACH CONDITION IS IN BOLD.

	Chikusei (Base model)			Pavia Centre (Base model)		
Condition	GSA	Base model fixed	Base model swap	GSA	Base model fixed	Base model swap
Nominal	43.20 / 2.73	58.15 / 0.72	–	34.78 / 5.23	50.63 / 1.90	–
Blur $\sigma = 1.5$	44.05 / 2.57	50.68 / 1.15	**57.97 / 0.73**	35.59 / 4.99	43.80 / 2.75	**50.56 / 1.90**
Blur $\sigma = 2.5$	42.77 / 2.82	54.26 / 0.88	**58.05 / 0.72**	34.36 / 5.36	47.30 / 2.14	**50.59 / 1.90**
Blur $\sigma = 3.0$	42.54 / 2.87	51.69 / 1.03	**57.89 / 0.74**	34.13 / 5.43	44.87 / 2.37	**50.52 / 1.91**
SRF shift -8 nm	42.86 / 2.91	**53.01 / 1.20**	52.54 / 1.31	34.58 / 5.36	49.46 / 1.99	**49.75 / 2.03**
SRF shift $+8$ nm	43.29 / 2.68	**53.03 / 1.19**	52.47 / 1.28	34.79 / 5.20	49.62 / 1.99	**49.82 / 2.02**
Noise, 40 dB SNR	43.04 / 2.85	**53.03 / 1.39**	–	34.71 / 5.37	**47.55 / 2.89**	–
Noise, 35 dB SNR	42.71 / 3.09	**49.11 / 2.20**	–	34.57 / 5.64	**44.36 / 4.19**	–
Noise, 30 dB SNR	41.84 / 3.67	**44.62 / 3.69**	–	34.16 / 6.37	**40.18 / 6.60**	–

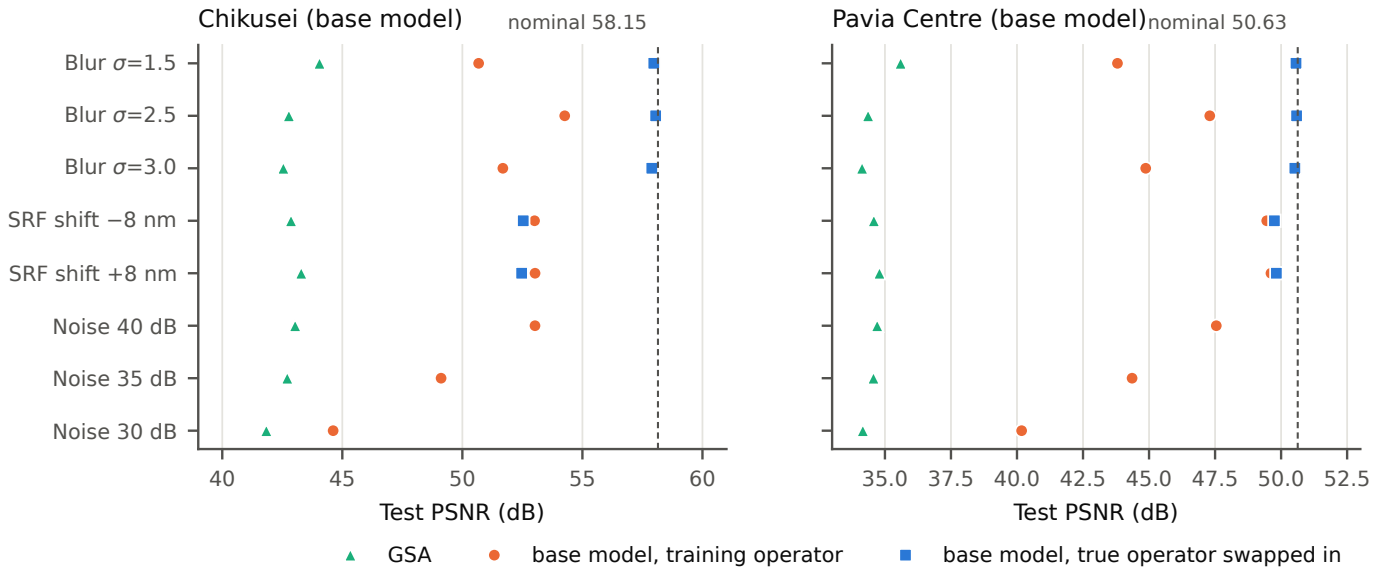


Fig. S3. Test PSNR under degradation mismatch and noise on Chikusei (left) and Pavia Centre (right). The dashed line is the nominal ExactFuse result.

TABLE S-III
Q2N ANALYSIS OF THE FINAL CHIKUSEI EXACTFUSE MODEL (SELF-ENSEMBLE, TEST SET)

Setting	Q2n
ExactFuse	0.9931
bands 1–8 (363–404 nm) set to ground truth	0.9988
bands 1–8 and 105–128 set to ground truth	0.9998
Deficit share, worst 10% / 50% of blocks	25% / 75%

S-V. FURTHER RESULTS OF THE LONG-TRAINED MODELS

Table S-IV lists the long-trained ExactFuse models of the comparisons with reported results, together with bicubic interpolation and GSA, under every PSNR and SSIM definition in use, and Table S-V gives the results of every test image. Chikusei images 5–8 are darker (their maximum is 0.36–0.47 of the dataset maximum), which lowers their strict SSIM but not their RMSE. Fig. S6 shows Chikusei image 7, the hardest test image, and Fig. S7 Pavia Centre image 3, the hardest there. The errors of bicubic interpolation and GSA follow every field boundary and road (mean absolute errors of 164.7 and 64.8 DN), whereas both ExactFuse models keep the error small and nearly uniform over the image (11.5 and 11.0 DN), apart from a few isolated pixels.

Fig. S8 shows the error of every band. ExactFuse is more accurate than GSA in all 128 Chikusei bands, and the final model improves on the first run in every band, most at the red edge (700–760 nm), where the mean band RMSE falls from 7.9 to 4.2 DN (from 9.8 to 1.1 DN at 724 nm). The error is largest in the first eight bands (363–404 nm), which lie below the first WorldView-2 band and therefore receive no high-resolution detail from the multispectral image, and in the last bands near 1000 nm, where the sensor's signal-to-noise ratio drops.

S-VI. COMPLETE BENCHMARK METRICS

Table S-VI lists all eight metrics of every method in the equal-budget benchmark; the per-job result files, including per-image values and training curves, are in the public repository.

S-VII. RELATED WORK WITH FULL REFERENCES

The main paper cites only the compared methods, the datasets and its closest references; this section gives its related work with all references, and the references of further details of the main paper.

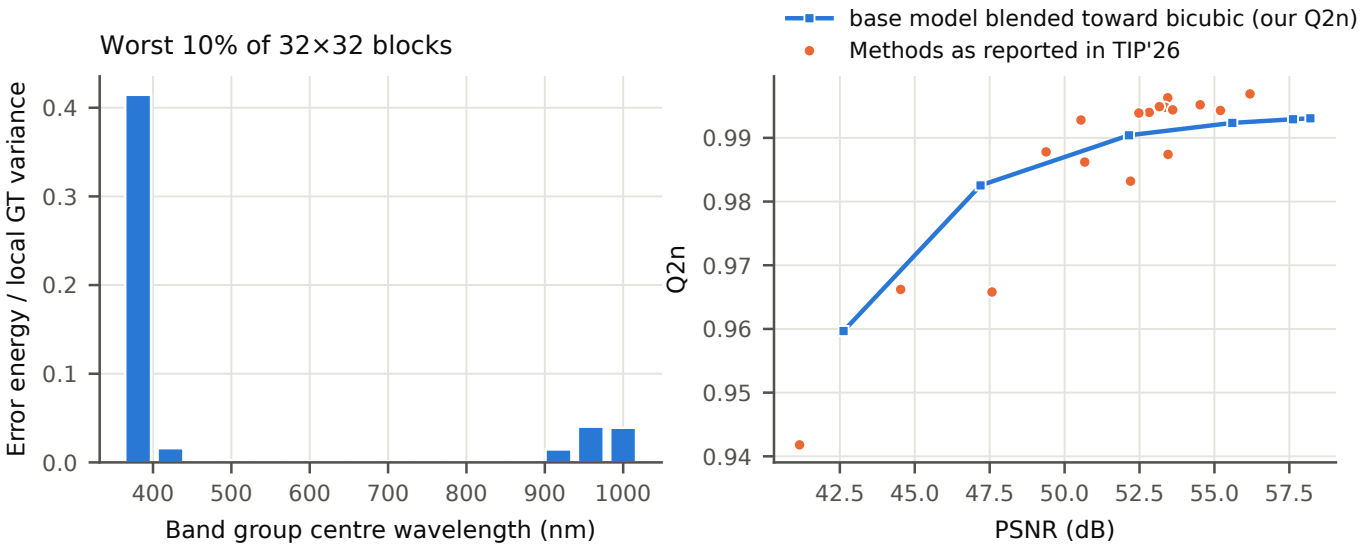


Fig. S4. Left: error energy relative to the local ground-truth variance per group of eight bands in the worst 10% of 32×32 blocks. Right: Q2n against PSNR when ExactFuse's output is blended toward bicubic interpolation (our implementation), and the (PSNR, Q2n) pairs reported in [1].

TABLE S-IV
THE LONG-TRAINED BASE MODELS AND THE CLASSICAL BASELINES UNDER ALL PSNR AND SSIM DEFINITIONS. P_g: PEAK = DATASET MAXIMUM (HEADLINE); P_i: PEAK = IMAGE MAXIMUM; P_b: BAND-WISE MEAN, PEAK 1. S_1: SSIM, DATA RANGE 1; S_i: DATA RANGE = IMAGE MAXIMUM.

Method	P_g	P_i	P_b	S_1	S_i	SAM	RMSE
Chikusei							
Bicubic	33.63	28.20	35.25	0.8500	0.7404	4.406	321.8
GSA [2]	43.20	37.76	44.82	0.9859	0.9778	2.730	107.7
Base model, first run	57.99	52.56	65.29	0.9989	0.9969	0.728	19.3
Base model	58.15	52.71	66.26	0.9989	0.9969	0.717	19.1
Base model, self-ensemble	58.20	52.77	66.33	0.9989	0.9970	0.713	18.9
Pavia Centre							
Bicubic	25.52	25.11	25.87	0.6427	0.6282	6.555	436.3
GSA [2]	34.78	34.37	35.19	0.9450	0.9438	5.229	157.9
Base model	50.63	50.22	51.11	0.9974	0.9970	1.895	23.7
Base model, ensemble	50.69	50.28	51.17	0.9974	0.9970	1.883	23.5

TABLE S-V
PER-IMAGE RESULTS OF THE FINAL MODELS (SELF-ENSEMBLE)

Image	PSNR	S_i	S_1	SAM	ERGAS	Q2n
Chikusei, Base model						
1	59.15	0.9994	0.9994	0.644	1.550	0.9957
2	60.45	0.9985	0.9994	0.643	0.972	0.9967
3	60.97	0.9987	0.9995	0.597	1.093	0.9962
4	57.10	0.9981	0.9988	0.862	1.334	0.9917
5	57.29	0.9951	0.9986	0.726	1.304	0.9929
6	57.51	0.9957	0.9987	0.669	1.357	0.9913
7	56.24	0.9953	0.9984	0.830	1.449	0.9913
8	56.90	0.9947	0.9986	0.729	1.584	0.9889
Mean	58.20	0.9970	0.9989	0.713	1.330	0.9931
Pavia Centre, Base model						
1	50.48	0.9977	0.9977	1.546	0.555	0.9987
2	52.30	0.9949	0.9964	2.722	0.813	0.9044
3	49.53	0.9979	0.9979	1.677	0.557	0.9991
4	50.45	0.9975	0.9976	1.588	0.543	0.9991
Mean	50.69	0.9970	0.9974	1.883	0.617	0.9753

A. Model-based fusion

Component substitution methods such as GSA [2] inject multispectral detail into the upsampled hyperspectral bands. Unmixing-based methods such as CNMF [3] factor both observations with shared endmembers, and subspace methods such as HySure [4] regularize the HR-HSI within a low-dimensional spectral subspace, whose dimension can be estimated with HySime [5]. FUSE [6] solves the resulting quadratic problem in closed form through a Sylvester equation. Pineau et al. [7] obtain an explicit solution by inverting the Hessian of a quadratic criterion under a linear mixing model, with circulant blurs. These methods use the forward model explicitly but rely on hand-crafted priors.

B. Supervised deep fusion

SSR-NET [8] and GuidedNet [9] learn a direct mapping from the two observations. Transformers model long-range spatial and spectral dependencies: PSRT [10] uses pyramid shuffle-and-reshuffle attention, MIMO-SST [11] multi-input multi-output supervision, LGCT [12] local–global collaboration, DSPNet [13] a dual spatial–spectral pyramid, SSDT [14] dilated attention, and MIMFormer [15], AMSF-Net [16], ASMNet [17] and DPFormer [18] multiscale spectral–spatial designs. Recent work also studies state-space models [19], [20], mixture-of-experts routing [21] and modality interaction [22]. General image-fusion frameworks such as U2Net [23] and OTPNet [24] have also been applied to this task. AMHF-Net [25] handles an arbitrary number of spectral bands. SA-RFM [26] performs non-rigid registration and fusion jointly with Riemannian flow matching.

C. Model-guided and unfolded networks

Algorithm unrolling [27]–[29] turns the iterations of an optimization method into network stages. MHF-Net [30], MoG-DCN [31] and DHIF-Net [32] unfold fusion-specific solvers, MHF-Net and DHIF-Net with operators learned as

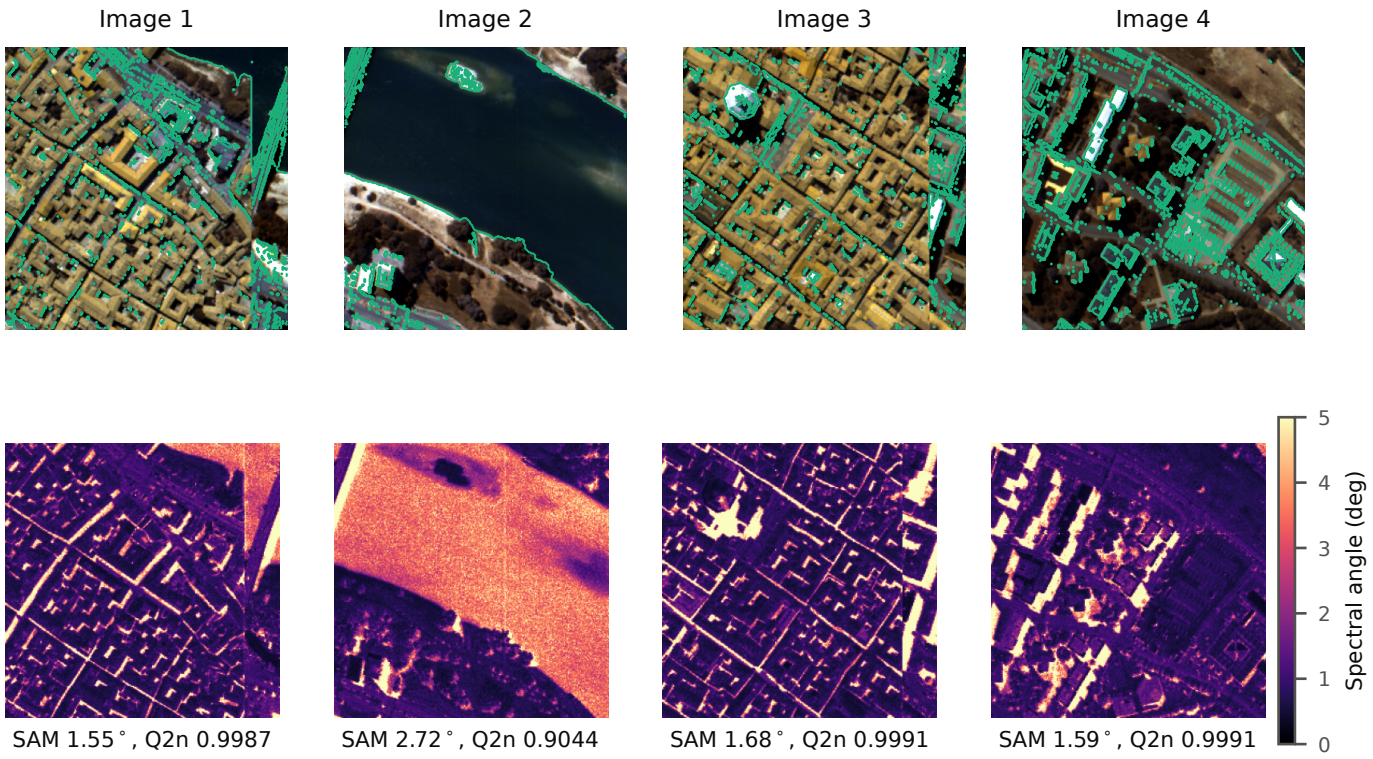


Fig. S5. The four Pavia Centre test images in false color (bands 57, 34 and 3; top), with water pixels (NDWI $>$ 0) outlined, and the spectral angle of ExactFuse's reconstruction at every pixel (bottom; color scale clipped at 5°).

convolution layers, DUNET [33] adds transformer priors for unregistered inputs, IR&ArF [34] unfolds the fusion of unregistered images at arbitrary resolution with a learned registration, HMPNet [35] extends unfolding to three modalities, and an information-aware transformer-based unfolding network (ITU-Net) [36], a mutual-guidance unfolding network (SMGU) [37], the perceptive spectral transformer unfolding network (PSTUN) [38], which also handles arbitrary scales, and the cross-domain-aware unfolding transformer CaFormer [39] strengthen the learned proximal steps with transformer priors. ExactFuse belongs to this family, and none of its prior's components is new: passing features from one stage to the next follows memory-augmented unfolding [40], spectral-wise attention inside an unfolded network appears in DAUHST [41], and the transposed attention is that of Restormer [42]. A single unfolded network that takes the blur kernel as an input, and therefore handles kernels it was not trained on, was shown for single-image super-resolution by USRNet [43].

D. Boundary conditions

In classical deconvolution, an image border handled differently by the model and by the data causes ringing that spreads inward, and reflexive (Neumann) boundary conditions, i.e., reflect padding, are a standard remedy [44]–[46]. In iterative reconstruction, a back-projection that is not the exact adjoint of the forward model changes the point to which the iteration converges [47], [48], and in convolutional networks zero padding is itself a source of spatial bias near the border [49]. In unfolded fusion networks the operators are either learned or implemented with the zero padding of convolution layers, while benchmark data are simulated with other border rules; to our knowledge, the effect of this mismatch has not been examined. Our contribution is on this side: an unfolded network whose operators, border included, are exactly those of the simulation, and an ablation that separates the exact operator from a zero-border and a learned operator under the same budget.

E. Generative fusion

Diffusion models have been conditioned on the two observations (DDIF [50]), constrained by the observation model during sampling (CDPS [51]), restricted to a learned subspace (ARGS-Diff [52]) or combined with differential attention in a two-stage conditioning scheme [1]. Null-space formulations [53] make the data-consistency part of the reconstruction exact. The two-stage conditional diffusion model of Chen *et al.* [1] holds the best published Chikusei $\times 4$ results in our survey and is our main reference; its comparison also includes the correlation-aware network CLSNet [54] and the dual cross transformer (DCTransformer) [55], denoted DCT in their table.

F. Further references of the main paper

The spectral responses are those of WorldView-2 [56] and IKONOS [57]; the operator estimate uses the relation of HySure [4], separable operators also appear in STEREO [58], and the true border rule of real images is unknown [59]. ExactFuse starts from the bicubic upsampling [60], runs its priors in mixed precision [61], uses a U-Net [62], deep supervision [63],

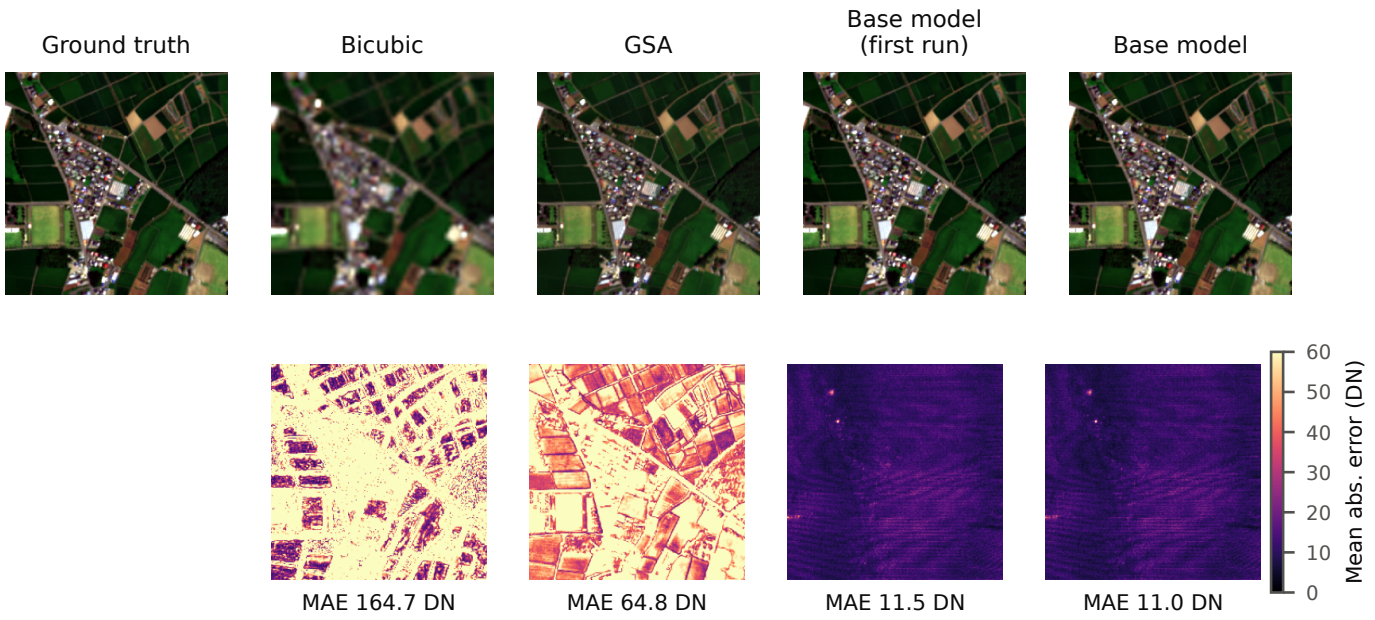


Fig. S6. Chikusei test image 7 in false color (bands 60, 40 and 20; top) and the mean absolute error over all bands in DN (bottom; color scale clipped at 60 DN).

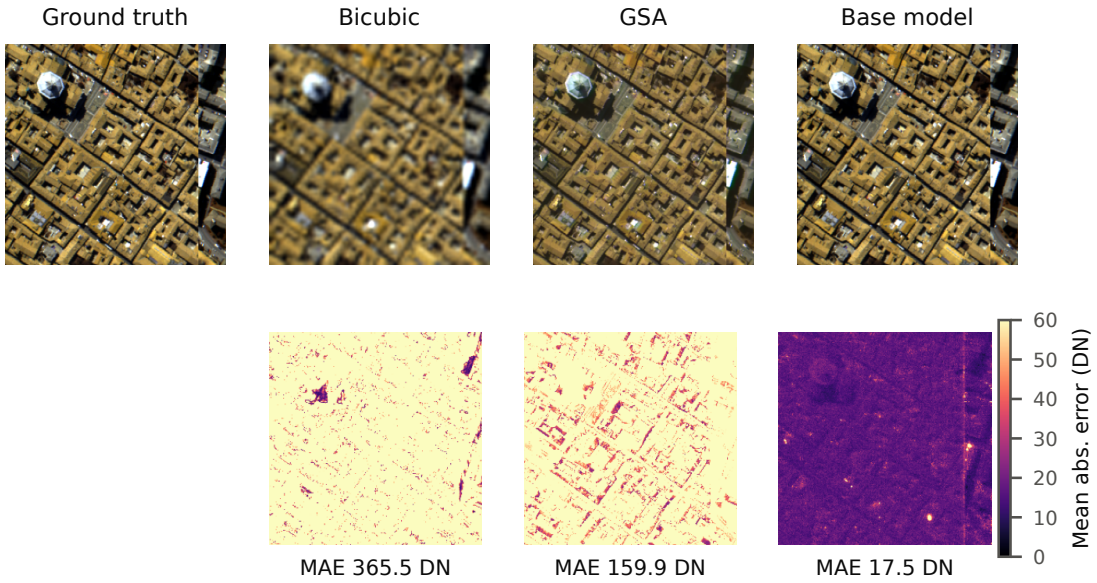


Fig. S7. Pavia Centre test image 3 in false color (bands 57, 34 and 3; top) and the mean absolute error over all bands in DN (bottom; color scale clipped at 60 DN).

FiLM modulation [64], a noise estimate after [65], randomized degradations as in blind super-resolution [66], [67] and the self-ensemble of [68]. Pairs follow Wald's protocol [69]. Training uses AdamW [70], cosine decay [71], distributed data parallelism [72] and, for some baselines, gradient checkpointing [73]; the ablation replaces the token mixer with depth-wise convolution [74] or window attention [75]. The metrics are SSIM [76], SAM [77], ERGAS [78], Q2n [79] as in the pansharpening toolbox [80], [81], and SCC [82]; significance uses the Wilcoxon signed-rank test [83]. The methods tabulated from [1] in the comparison with reported results are GSA [2], CNMF [3], FUSE [6], U2Net [23], PSRT [10], DCT [55], DDIF [50], MIMFormer [15], MIMO-SST [11], OTPNet [24], SMGU [37], AMSF [16], SSDT [14], ASMNet [17], DPFormer [18] and CLSNet [54]. CLoRF [84] reaches arbitrary output resolutions through a self-supervised continuous low-rank factorization; PIDM [85] learns degradations with non-uniform lens aberrations, which lie outside the separable operators used here; and an early sensor-simulation method [86] already lets each multispectral band add detail only to the hyperspectral bands within its spectral response.

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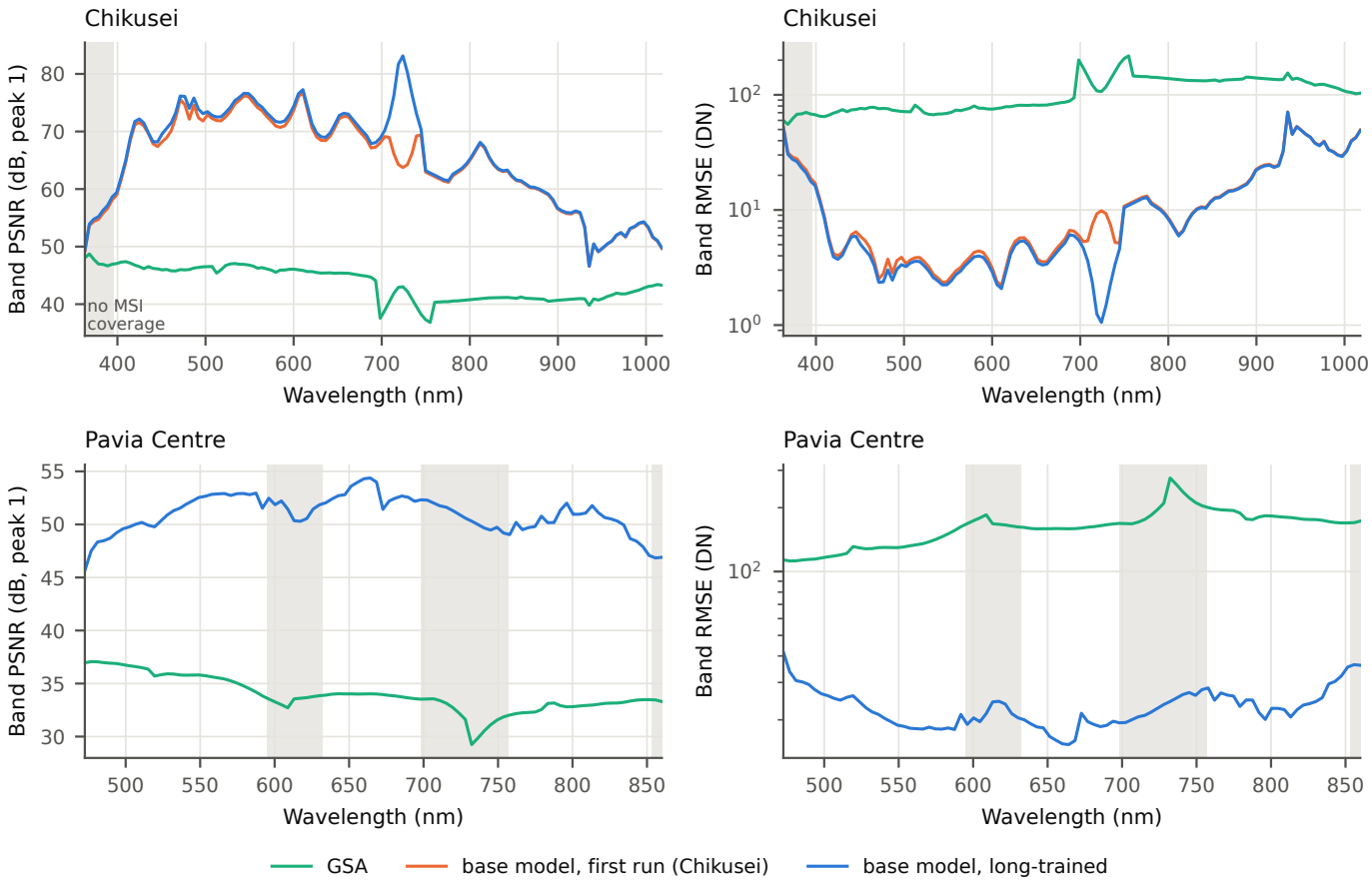


Fig. S8. Band-wise PSNR (left, peak 1) and RMSE in DN (right, log scale) over the test images of Chikusei (top) and Pavia Centre (bottom). Shaded ranges have no multispectral coverage: below the first WorldView-2 band for Chikusei, and between and beyond the IKONOS bands (50% edges) for Pavia Centre.

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TABLE S-VI
ALL METRICS OF THE EQUAL-BUDGET BENCHMARK (SINGLE PASS; SSIM WITH THE DATA RANGE SET TO EACH IMAGE'S MAXIMUM; RMSE IN DN; –: RUN NOT AVAILABLE).

Method	PSNR	SSIM	SAM	ERGAS	CC	SCC	Q2n	RMSE
Chikusei ×4								
Bicubic	33.63	0.7404	4.406	9.011	0.8408	0.7648	0.6914	321.81
GSA	43.20	0.9778	2.730	3.279	0.9905	0.9922	0.9596	107.69
SSR-NET	55.55	0.9958	0.954	1.707	0.9948	0.9987	0.9901	25.30
PSRT	53.99	0.9948	1.057	1.794	0.9942	0.9982	0.9887	30.57
DCT	53.18	0.9945	1.133	2.066	0.9930	0.9978	0.9871	33.58
MIMFormer	56.96	0.9964	0.801	1.488	0.9960	0.9989	0.9918	21.65
DHIF-Net	54.39	0.9953	1.020	1.660	0.9951	0.9983	0.9900	29.37
U2Net	53.35	0.9943	1.076	1.948	0.9937	0.9977	0.9863	32.92
MIMO-SST	57.27	0.9965	0.780	1.419	0.9963	0.9990	0.9922	20.90
OTPNet	57.06	0.9965	0.786	1.448	0.9962	0.9990	0.9922	21.39
SMGU-Net	55.14	0.9957	0.949	1.705	0.9947	0.9986	0.9898	26.60
CLSNet	56.65	0.9963	0.811	1.514	0.9959	0.9989	0.9917	22.35
CaFormer	53.56	0.9949	1.054	2.138	0.9926	0.9980	0.9880	32.07
M2DTN	54.87	0.9952	1.003	1.977	0.9931	0.9987	0.9877	27.44
ExactFuse	58.05	0.9969	0.724	1.334	0.9967	0.9992	0.9930	19.21
Pavia Centre ×4								
Bicubic	25.52	0.6282	6.554	9.852	0.8174	0.7526	0.6806	436.32
GSA	34.78	0.9438	5.229	3.494	0.9958	0.9922	0.9276	157.93
SSR-NET	48.45	0.9954	2.443	0.758	0.9991	0.9991	0.9719	30.46
PSRT	47.87	0.9958	2.212	0.801	0.9990	0.9989	0.9716	32.60
DCT	46.94	0.9953	2.442	0.888	0.9988	0.9985	0.9712	36.26
MIMFormer	49.63	0.9967	1.982	0.674	0.9993	0.9993	0.9738	26.66
DHIF-Net	47.67	0.9956	2.295	0.840	0.9989	0.9988	0.9704	33.42
U2Net	47.29	0.9955	2.327	0.858	0.9989	0.9987	0.9708	34.96
MIMO-SST	49.69	0.9967	1.961	0.670	0.9993	0.9993	0.9741	26.48
OTPNet	50.08	0.9968	1.951	0.653	0.9993	0.9994	0.9737	25.29
SMGU-Net	48.03	0.9958	2.331	0.783	0.9990	0.9990	0.9723	32.05
CLSNet	49.50	0.9967	1.993	0.684	0.9993	0.9993	0.9735	27.09
CaFormer	48.43	0.9961	2.145	0.757	0.9991	0.9991	0.9724	30.56
M2DTN	48.65	0.9963	2.156	0.729	0.9992	0.9991	0.9727	29.76
ExactFuse	50.63	0.9970	1.892	0.619	0.9994	0.9995	0.9753	23.70

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